

01 The Paradigm Shift: Rejecting the Fantasy User Interface

SECTION

How Phil Lord, Christopher Miller, and Production Designer Charles Wood inverted Hollywood's green-screen convention for tactile aerospace realism.

For over two decades, blockbuster science fiction cinema relied heavily on the Fantasy User Interface (FUI)—characterized by transparent holographic displays, mid-air gestural swiping, floating geometric arrays, and kinetic non-functional motion. While visually flashy, FUIs detach the audience and actor from physical engineering realities. For the \$250M Amazon MGM Studios adaptation of Andy Weir's *Project Hail Mary*, the filmmakers completely rejected this convention, building a 100% practical, operational spaceship interior where all displays were live, physical HDMI-driven screens operated on set by Compuhire and Studio C Design.

FUI vs. Practical Aerospace UX Metrics

Evaluating functional performance across set engineering, actor performance, and post-production metrics.



Data mapped from Studio C Design & Compuhire production post-mortems.

Key Paradigm Contrast

Attribute	Standard Sci-Fi (FUI)	Project Hail Mary
Display Surface	Glass / Blank Green Screen	Ruggedized High-Contrast LCDs
Interaction Type	Pantomimed Air Gestures	Physical Switches, Dials & Touch
Light Emission	Green Spill (Requires Rotoscoping)	True-Color Diegetic Ambient Light
Inspiration Base	Pure Graphic Fantasy	ISS, Soyuz, Boeing 747 Cockpit
Playback Execution	Post-Production VFX Compositing	Live On-Set Microcontrollers / HDMI

Design Insight: The interior architecture relied heavily on high-density commercial aviation layouts (Boeing 747-400 switchgear) combined with rugged Russian Soyuz space capsule aesthetics, ensuring Ryan Gosling was reacting to authentic telemetry rather than imaginary markers.

02 On-Set Hardware Integration & Gimbal Playback System

SECTION

The mechanical and electronic infrastructure powering a rotating, hydraulic 8-foot capsule set.

Building a fully operational, live-screen spacecraft cockpit presented monumental hardware engineering challenges. The 8-foot wide capsule was mounted on a multi-axis kinetic shudder gimbal simulating centripetal gravity and violent thrust. Compuhire technicians engineered modular power, data routing, and hardware microcontrollers directly into the set framework to ensure uninterrupted, zero-latency screen updates during violent physical movement.

Interactive On-Set Signal Pipeline

Click on any subsystem component in the pipeline below to inspect its operational spec and hardware role.

1. Physical Switchgear (MDBs)
2. Media Players (Neumi 4K)
3. Slip-Ring Gimbal Axis
4. High-Contrast LCD Displays
5. Off-Set Cloud Pipeline

Microcontroller Development Boards (MDBs)

HARDWARE INPUT TIER

Custom-wired Microcontroller Development Boards (MDBs) were embedded directly behind physical switches, heavy mechanical pushbuttons, and industrial potentiometers. When Ryan Gosling depressed a physical toggle, the MDB instantly translated the physical contact into high-speed control signals sent directly to the local video engines.

LATENCY:

< 12ms Response

PROTOCOL:

Custom Serial/UDP

REDUNDANCY:

Dual-Bus Architecture

MOUNTING:

Anti-Vibration Rated

Gimbal Motion Cabling

Specialized high-durability HDMI and power cabling routed through multi-channel slip-rings along the central rotation axis of the gimbal to prevent twisting during 360° set spinning.

Neumi Atom 4K Array

Ultra-compact 4K solid-state media engines provided reliable, shock-resistant video playback that couldn't crash or skip frame buffers during intense hydraulic shudder effects.

Live Cloud Sync Pipeline

Studio C graphics designers pushed rapid code updates and typographic edits from editing bays directly to the set playback servers in minutes via cloud syncing workflows.

03 Astrophage Spacecraft Telemetry & Dark UI Simulator

SECTION

Interactive demonstration of Studio C Design's high-contrast, low-eyestrain fuel and thermal diagnostic dashboard.

Astrophage—the biological energy source powering the *Hail Mary*—is volatile and exponentially hot. To monitor this alien energy fuel, Studio C designed a strict Dark UI framework. High-contrast typography, subtle glassmorphism, and dynamic thermal gradients preserved the astronaut's night vision while instantly signaling life-critical system states.

● HAIL MARY TELEMETRY // NOMINAL CRUISE

ASTROPHAGE DRIVE ENGINE DIAGNOSTIC

Nominal Cruise Thermal Spike Critical Warning

FUEL MASS (PETAGRAMS)

2.14 PG

Consumptions Nominal

THERMAL CELL TEMP

96.4 °C

Threshold: 1400 °C

CONTAINMENT FIELD

99.8 %

Harmonic Lock Active

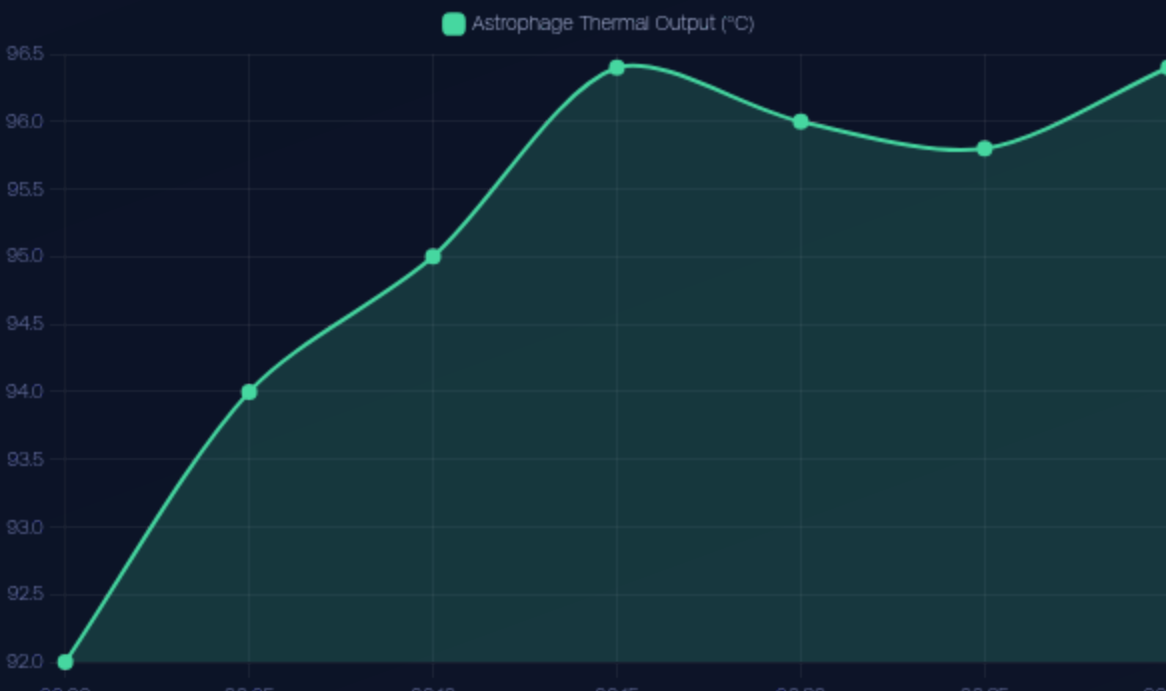
OXYGEN SCRUBBER

21.2 kPa

Atmosphere Nominal

REAL-TIME ASTROPHAGE ENERGY CURVE

Live Stream: 100ms Telemetry



Cognitive Load Spatial Hierarchy: Essential flight vectors and containment safety reside in the astronaut's primary line of sight. Secondary long-term orbital mechanics and atmospheric scrubbing metrics are pushed to peripheral displays. If an error spikes in the periphery, ambient lighting changes color dynamically to redirect the actor's eye naturally.

04 Diegetic Lighting & Cooke /i Metadata Integration

SECTION

How DP Greig Fraser utilized real screen emission and frame-accurate lens metadata to revolutionize VFX defocus.

When an actor stands in front of a green screen, artificial green light bounces onto skin and clothing ("green spill"), destroying visual authenticity and requiring laborious post-production cleanup. By shooting with practical screens, Director of Photography Greig Fraser allowed the actual high-contrast screen displays to cast real, shifting light on Ryan Gosling's face.

Cooke /i Lens Metadata Visualizer

Adjust the simulated ARRI ALEXA 65 camera parameters to see how real-time lens metadata drives exact optical defocus on exterior space VFX.

Aperture (T-Stop): T2.8

Focus Distance (Foreground Screen vs Background): 1.2 meters (Screen Sharp)

Anamorphic Focal Length: 65 mm (IMAX 143:1 crop)

COMPARED TELEMETRY: OK

SCREEN LIGHT CAST: 450 LUX

How Metadata Bridges Practical & VFX

Every frame shot on the ARRI ALEXA 65 recorded exact frame-by-frame metadata via Cooke /i Technology. This "love letter" from DP Greig Fraser to visual effects houses (ILM & Framestore) eliminated the need to guess depth of field or lens distortion.

Focus Distance Mapping

Post-production compositing tools automatically match the optical blur of exterior CGI starfields to the exact depth-of-field falloff created by the practical screen glass.

Photometric Light Matching

The light illuminating Ryan Gosling's skin matched the color spectrum and intensity of the telemetry graphics, making the actor feel tangibly integrated into the spacecraft.

Elimination of Green Spill Cleanups

Zero rotoscoping artifacts or green fringe matting allowed natural hair edges, glass reflections, and suit details to remain untouched.

05 The Eridian Interspecies HCI Lab (Rocky & Grace)

SECTION

Simulating the optical-to-acoustic hardware converter and Python acoustic-chord translation layer.

The alien engineer "Rocky" (an Eridian) has no visual organs, perceiving the universe entirely through acoustic echolocation. Because human screens are visually useless to Rocky, he builds a mechanical hardware converter that scans 2D optical display patterns and renders them into 3D tactile/acoustic impulses. Concurrently, Ryland Grace writes a Python software interface to translate Eridian acoustic chords into English text.

ACOUSTIC / OPTICAL INTERPOLATION PIPELINE

Human Visual Interface <=> Eridian Echolocation Converter

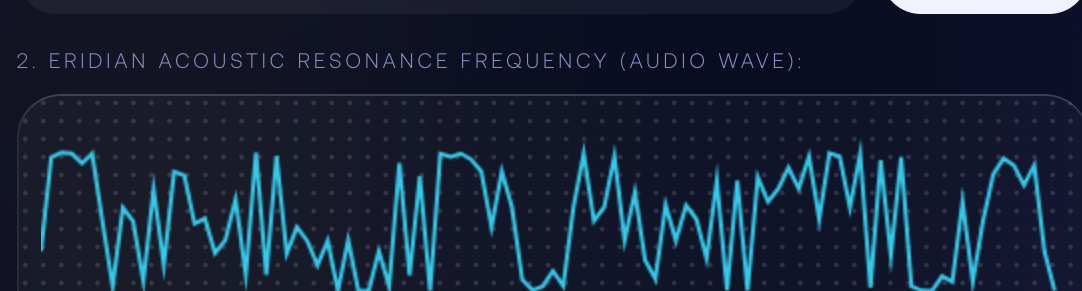
Python 3.12 / PyAudio Signal Engine

1. HUMAN PHRASE / SYSTEM STATUS INPUT:

Astrophage Fuel Nominal

Synthesize

2. ERIDIAN ACOUSTIC RESONANCE FREQUENCY (AUDIO WAVE):



3. ERIDIAN CONVERTER TACTILE / CHORD OUTPUT:

// Waiting for acoustic interpolation sequence...

Converter Mechanism: Reads 2D light pixels -> Converts to micro-vibrational pitch chords

Philosophical UX Principle: True interface design does not force an alien user to conform to human sensory limits. Instead, it provides an empathetic translation layer that preserves the natural cognitive architecture of both species.

06 Production Economics & Visual Effects Budget Allocation

SECTION

Analyzing the economic return on front-loaded practical hardware investments.

While building a working physical spaceship with embedded microcontrollers required heavy capital expenditure during pre-production, it dramatically reduced post-production visual effects costs. By capturing diegetic displays in-camera, Amazon MGM Studios eliminated thousands of tedious rotoscoping hours, freeing VFX vendors like ILM and Framestore to focus exclusively on high-value exterior space sequences.

Post-Production Workload Comparison

Estimated VFX labor distribution: Standard Green Screen vs Practical On-Set Playback.



Economic Takeaways

2,000+

VFX Shots Saved from 2D Screen Rotoscoping

100%

Practical Interactive Light Capture

Grounded Performance Dividends: Beyond cost savings, the tangible environment enabled Ryan Gosling to perform with visceral authenticity. Rather than pausing to wait for cues on a blank green screen, pressing live buttons produced instantaneous set reactions, enhancing emotional realism.